

Simon Chen

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EDUCATION

Boston University

Boston, MA

Bachelor of Arts in Computer Science

Graduated: Spring 2026

- Relevant Coursework: Data Structures & Algorithms, Operating Systems, Distributed Systems, Databases.

EXPERIENCE

Technical Director / Software Engineer

December 2025 – Present

Ezesports

New York, NY

- Migrated an esports league platform from Google Sites to Next.js with a Supabase backend, replacing manual content updates with live scoring and automated season management.
- Designed a PostgreSQL schema on Supabase using Drizzle ORM, supporting 500+ matches across 20+ teams, and built RESTful endpoints for player statistics, standings, and historical data.
- Cut page load time by 75% (from 3.2s to 0.8s) by implementing edge caching on Cloudflare, route-based code splitting, and asset optimization for the Next.js frontend.

Software Engineering Intern

February 2026 – June 2026

RESET Standard

Shanghai, China

- Doubled the platform's supported environmental data categories (from 2 to 4) by independently implementing end-to-end water and waste data integration, developing Rails backend ingestion pipelines, PostgreSQL schemas, and client-facing React/Vite/Tailwind dashboards.
- Engineered asynchronous data pipelines with RabbitMQ, Redis, and Sidekiq to process long-running ingestion jobs in the background, decoupling external API calls from user-facing requests.
- Implemented a key client-requested feature by redesigning project-device mapping workflows, refactoring legacy forms to streamline the onboarding of environmental monitoring devices.

IT Consultant

January 2024 – July 2025

Boston University Engineering IT

Boston, MA

- Automated software deployment for 100+ devices using Microsoft Deployment Toolkit and created custom USB deployment solution for Windows 11, reducing deployment time by 40%
- Managed 200+ user accounts and permissions via NAS, conducting access audits and boosting system uptime by 15%
- Enhanced endpoint security with CrowdStrike for 50+ devices, resolving 20+ tickets/week with 100% incident resolution rate

PROJECTS

Hermes Letters | *Next.js, TypeScript, Supabase, Drizzle ORM, PostgreSQL*

June 2026

- Built a private long-form messaging platform for a group of 20+ friends during a study abroad semester, delivering 30+ letters.
- Ensured consistent letter claims via atomic SQL updates and secured sessions in XSS-resistant httpOnly cookies.
- Derived user connections at query time from kept-letter records using indexed lookups, avoiding a separate relationship table and associated database sync issues.
- Hardened media storage by validating uploads via server-side magic-byte signatures, storing files in a private Supabase bucket, and restricting delivery to 1-hour signed URLs.

Persephone | *Python, Cython/C++17, Rust, Modal, FAISS*

May 2026

- Built and operated a production Kalshi BTC prediction-market trading system that generated \$8K net PnL and traded \$200K volume in its first 4 live days, owning market-data ingestion, theo serving, strategy evaluation, exchange execution, risk gates, and operator controls.
- Engineered a GBM theo layer to replace KDE inference, reducing model query latency from ~1.5ms to ~20μs per query (~75x), while preserving the live theo interface used by strategy evaluation, cache publication, and operator tooling.
- Reduced backtest and parameter-sweep iteration from ~14 hours on local MacBook CPUs to ~15 minutes on Modal NVIDIA GPUs by moving diagnostics, tick backtests, and sweeps onto cloud workers with FAISS GPU search and CuPy-vectorized math.
- Optimized KDE serving with cached FAISS IVF indexes and FAISS thread pinning, avoiding OpenMP oversubscription and cutting 155K-row neighbor search from 758us p50 flat FAISS to 154us p50 with IVF.

TECHNICAL SKILLS

Languages: C++17, C/C++, Cython, Go, HTML/CSS, Java, JavaScript, Python, Ruby, Rust, SQL, Solidity

Frameworks & Libraries: Boost Libraries, CuPy, DBSCAN, Django, Drizzle ORM, Express.js, FAISS, FastAPI, Folium, Next.js, Node.js, OpenAI Whisper

Databases & Infrastructure: AsyncStorage, Auth0, CME MDP 3.0, Circom, Cloudflare, Cloudflare Pages, Cloudflare Tunnels, Docker, Docker Compose, Flyway migrations, GeoPandas, Git

Platforms & Systems: AWS, CrowdStrike, Ethereum, Heroku, Kalshi, MetaMask, Microsoft Deployment Toolkit, Modal, NAS, ServiceNow, Thirdweb SDK